Cove — Full Usability & Functionality Spec

Oct 30, 2026 · @Qadeer

Overview

Cove is a Windows tray app that puts file tools and desktop status into one pill at the top-center of the screen. It stays hidden until you call it with a shortcut, or until something worth showing happens.

Design principles:

- **Invisible by default.** Nothing is on screen until you call Cove or an event needs you.
- **Keyboard first.** Every feature is reachable from the keyboard; the mouse is optional.
- **Polite with focus.** Alerts never take focus from the window you are typing in. Panels you open with a shortcut take focus, then hand it back when they close.
- **Never destroys data.** Originals are never overwritten, moves can be undone, deletes go to the Recycle Bin.
- **Modular.** Every module can be switched off without affecting the others.

Module	Group	What it does	How you reach it
Files panel	Files	Takes in files and runs actions on them	Ctrl+Alt+D, Cove key + D
Convert	Files	Changes a file's format	Files panel, right-click menu, alerts
Compress	Files	Shrinks a file to a target size	Files panel, right-click menu, alerts
Upload	Files	Uploads to OneDrive or Google Drive and copies a link	Files panel, right-click menu, alerts
Copy text	Files	Reads text out of images and PDFs (OCR)	Files panel, right-click menu, alerts
Shelf	Files	Holds files temporarily while you navigate	Ctrl+Alt+S, Cove key + S
Auto-Sort	Files	Files new downloads into folders by rules	Automatic

Module	Group	What it does	How you reach it
Bulk tools	Files	Rename, strip metadata, merge or split PDFs, zip	Files panel, right-click menu
Media	Desktop	Now playing and playback controls	Automatic, Ctrl+Alt+M
Mixer	Desktop	Per-app volume and output devices	Ctrl+Alt+M (Mixer tab)
Mic and Camera	Desktop	In-use indicator and global mute	Automatic, Ctrl+Alt+A
Devices and Battery	Desktop	Connect alerts and low-battery warnings	Automatic, Cove key + B
Timer and Focus	Desktop	Timers, Pomodoro, quiet focus sessions	Ctrl+Alt+T, Cove key + T

The pill

The pill lives at the top-center of the main monitor, 8 px below the screen edge, and moves between five states. Sizes below are starting points to tune by eye.

State	Approx. size	What it shows	Opens when	Closes when
Hidden	Nothing	Nothing (optional faint 2 px hint line)	Default	A shortcut or an event
Peek	200 x 36 px	One status line: song or clock, mic dot, job spinner, timer countdown, shelf count	A job is running after a panel closes, or optional hover	3 s with nothing to show, or Esc
Alert	360 x 56 px	Icon, one-line message, up to two buttons	An event fires	After 4 s (8 s for urgent), a button, or Esc
Menu	360 x 64 px	Cove key letters and a command field	Ctrl+Alt+Space	A letter, Esc, or a click outside

State	Approx. size	What it shows	Opens when	Closes when
Panel	520 px wide, 240 to 420 px tall	A full module: Files, Media, Timer, Shelf, Devices, Jobs	A direct shortcut or a menu letter	Esc, the same shortcut again, or a click outside (unless pinned)

Movement. The pill slides down and grows into the next state in about 250 ms, and shrinks back up in about 180 ms. With reduced motion on, it only fades.

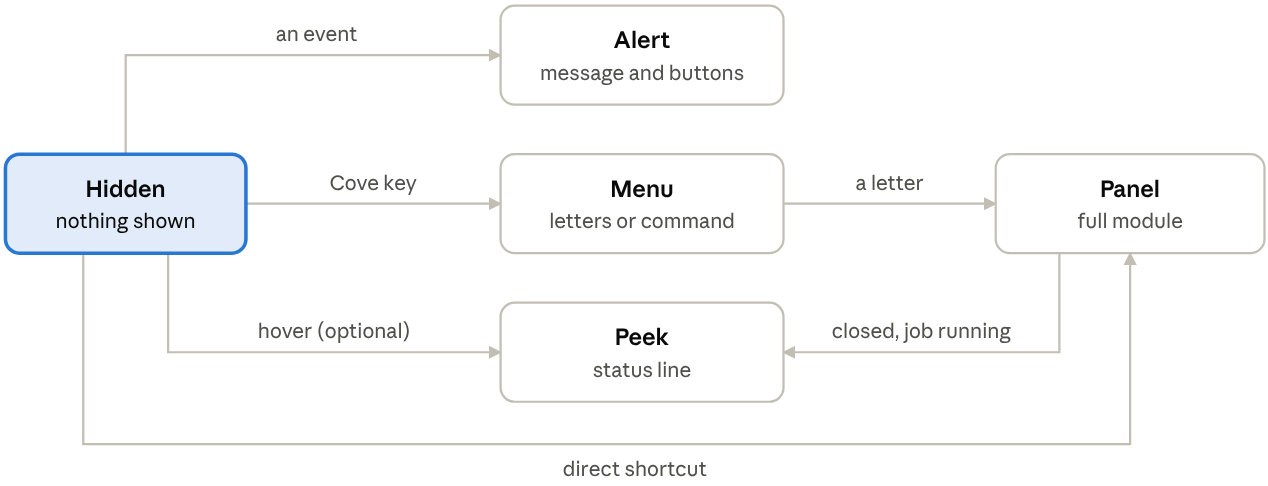
Focus. Peek and Alert never take keyboard focus, so typing continues in your current window. Menu and Panel take focus so you can type and press number keys, then return it to the previous window when they close.

Mouse. Peek lets clicks pass through it except on its icons. Alert buttons are clickable. A pin button in the panel's corner keeps it open while you click elsewhere.

Fullscreen. While a fullscreen game, video or presentation is in front, the pill stays hidden and non-urgent alerts wait (see Automatic pop-ups).

Look. It follows the Windows light or dark theme and accent colour, uses a Mica or acrylic background, and has fully rounded ends.

The pill rests hidden and opens only on a shortcut or an event



Esc, a click outside, or a timeout returns any state to Hidden.

Pill states · five states and what moves between them

Shortcuts and events are the only ways out of Hidden by default; hover is an opt-in extra.

Ways to open Cove

There are six ways in, and the first two cover almost everything. Pressing the same shortcut again closes what it opened; Esc closes any panel.

1. **Cove key (Ctrl+Alt+Space).** Opens the Menu: one letter jumps to any panel, or type a command.
2. **Direct shortcuts.** One shortcut per feature, for the ones you use most (table below).
3. **Automatic alerts.** Events open an Alert on their own: a song change, a finished download, a job done, the mic turning on.
4. **Tray icon.** Left-click opens the Menu. Right-click shows Files, Shelf, Timer, Pause alerts, Settings and Quit.
5. **Explorer right-click menu.** Cove > Convert, Compress, Upload and more, with progress shown in the pill.
6. **Hover (optional, off by default).** Resting the cursor for 150 ms in a 400 x 4 px strip at the top-center opens Peek. It never triggers while a window is being moved, so Windows Snap Layouts keeps working.

Default shortcut	Opens or does
Ctrl+Alt+Space	Cove key Menu
Ctrl+Alt+D	Files panel
Ctrl+Alt+M	Media and Mixer panel
Ctrl+Alt+T	Timer panel
Ctrl+Alt+S	Shelf panel
Ctrl+Alt+A	Mute or unmute the mic

Every shortcut can be changed or switched off. If another app already owns a combination, Windows refuses it; Cove then names the conflict and asks for a different one.

Cove key menu and command bar

Ctrl+Alt+Space opens a small Menu that shows its own letters, so nothing has to be memorised. A letter jumps straight to a panel; a slash switches to typing a command.

Letter	Opens
D	Files panel
M	Media and Mixer
T	Timer and Focus
S	Shelf
B	Devices and Battery
J	Jobs (running and recent)
A	Toggles mic mute, then closes
, (comma)	Settings

Command mode. Press / in the Menu and type. Suggestions appear as you type, Tab completes, Enter runs, and the up arrow brings back your last command.

Command	Example	Does
convert	convert pdf	Converts the current files to that format
compress	compress 10mb, compress discord	Compresses to a size or preset
upload	upload	Uploads and copies a link
text	text	Copies text out of an image or PDF
shelf	shelf, shelf clear	Adds the current files to the Shelf, or clears it
timer	timer 10, timer 1h30	Starts a timer
focus	focus 25	Starts a focus session
media	play, pause, next, prev	Controls playback
vol	vol 40, vol spotify 20	Sets system or per-app volume
mute	mute, unmute	Mic mute
settings	settings shortcuts	Opens Settings at a section

Which files a command uses. First the files selected in the Explorer window you were in, then files copied in the last 2 minutes. If there are none, the Files panel opens so you can drop some.

Files panel

Ctrl+Alt+D opens the Files panel, which loads your files from wherever they already are, so dragging is optional. Because the panel is on screen before any drag starts, dropping onto it always works.

Getting files in, in the order Cove checks:

1. **Explorer selection.** Files selected in the Explorer window you were using load automatically. The header reads "3 files from Downloads".
2. **Recently copied files.** Files copied with Ctrl+C in the last 2 minutes appear as a chip: "Use 2 copied files".
3. **Drag and drop.** Drop files or a folder on the drop area; it highlights while a file is over it. A folder adds the files inside it (top level only, with an option to include subfolders).
4. **Browse.** A standard Windows file picker.
5. **From the Shelf.** An "Add from Shelf" button.

You can keep adding files. Each card has an X to remove it, and Clear empties the list.

File cards. Each file shows a thumbnail or icon, name, type, size and folder. Images also show dimensions; videos and audio show duration.

Layout, top to bottom: a header with the file count and a pin button, the file cards, an action row (1 Convert, 2 Compress, 3 Upload, 4 Copy text, 5 Shelf, 6 More), the selected action's options, and a Run button.

Mixed file types. Cove offers only the actions that work for every file in the list. Convert then shows target formats grouped by type, such as images to JPG and videos to MP4.

Keyboard. Number keys pick an action, arrow keys move between options, Enter runs, Delete removes the selected card, Esc closes the panel without stopping any running job.

File actions

Every action writes a new file next to the original and never overwrites anything. If the name is taken, Cove adds a number, such as "photo (1).jpg".

Convert

From	To	Engine
Images: PNG, JPG, WebP, HEIC, BMP, GIF, TIFF	PNG, JPG, WebP, ICO, PDF	ImageMagick (with HEIC support)
Video: MP4, MOV, MKV, AVI, WebM	MP4, WebM, GIF, MP3 (audio only)	ffmpeg
Audio: MP3, WAV, M4A, FLAC, OGG	MP3, WAV, M4A, FLAC	ffmpeg
Documents: DOCX, PPTX, XLSX, ODT	PDF	LibreOffice (headless)
PDF	PNG or JPG, one image per page	Ghostscript or a PDF library

Options: a quality slider for JPG, WebP and MP3; an optional resize for images; start time, end time, frame rate and width for GIFs. Output goes to the same folder by default, or a fixed folder set in Settings.

Compress

- **Presets:** Discord (10 MB), Email (25 MB), Web image (about 500 KB), or a custom size or quality.
- **Video:** Cove works out the bitrate that fits the target size for that duration, then lowers the resolution if the bitrate would look too poor.
- **Images:** quality steps down, then size, until the file is under the target.
- **PDFs:** embedded images are downsampled.
- **Impossible targets:** if a target can't be reached at watchable quality (a 2-hour video into 10 MB), Cove warns first and shows the best size it can reach.
- Output is named "name (compressed).ext". An option copies the result to the clipboard as a file, ready to paste into Discord or an email.

Upload

- Providers: OneDrive and Google Drive, signed in once in Settings through the browser.
- Files go into a "Cove" folder in your cloud storage.

- Link access: anyone with the link can view (default), only your organisation, or no link (upload only).
- Several files: one link each, or zipped into one link (setting).
- Large files upload in resumable chunks with a progress bar.
- When it finishes, the link is on your clipboard and an Alert offers Open and Copy again.

Copy text

- Reads text from images and PDF pages with Windows' built-in OCR, in the languages installed on your PC.
- The text is copied right away. The panel shows a preview with Copy again and Save as .txt.

Send to Shelf

Adds the files to the Shelf (see Shelf).

Bulk tools (under More)

- **Rename:** a pattern such as "Lab_{n}" or "{date}_{name}", with a live before-and-after preview.
- **Strip metadata:** removes location, camera and author data. It makes clean copies by default; changing the originals needs a confirmation.
- **Merge PDFs:** drag the cards into order, then merge into one PDF.
- **Split PDF:** extract page ranges such as "1-3, 7".
- **Zip:** packs the files into one .zip.

Jobs

Every action runs as a background job, so closing the panel never stops the work. A job moves through five stages:

1. **Queued.** It appears in the panel with its file names. Two jobs run at once by default; the rest wait their turn.
2. **Running.** A progress bar shows percent and time left, or a spinner when the length is unknown. Video jobs run at low priority so the PC stays responsive.

3. **In the background.** If you close the panel, Peek shows a spinner and a count ("2 jobs"). Cove key + J opens the Jobs list.
4. **Done.** An Alert shows "Saved Essay.pdf" with Open, Show in folder and Copy. Uploads add Copy link.
5. **Failed.** An Alert gives the reason in plain words, with Retry and Details.

Cancel. The X on a job, in the panel or the Jobs list, stops it and deletes any half-written file.

Jobs list. Shows running jobs and the last 20 finished ones with their results; clicking a result opens it.

Failure	What the Alert says	What Cove offers
File locked by another app	Couldn't convert: Essay.docx is open in Word	Retry after closing it
Not enough disk space	Not enough space on C: (needs about 1.2 GB)	Choose another output folder
Unsupported or damaged file	Can't read this file as a video	Details
No internet	Upload paused: no connection	Retries 3 times, then Retry
Cloud sign-in expired	Sign in to OneDrive again	Opens sign-in
Compression target unreachable	Smallest possible is 18 MB	Use 18 MB, or cancel

Automatic pop-ups

Twelve events can open an Alert on their own. Each can be set to Alert, Peek only (just updates the status line) or Off.

Event	Example message	Buttons	Default
Mic starts being used	Mic on: Discord	Mute	Alert

Event	Example message	Buttons	Default
Timer finishes	Timer done (25 min)	+5 min, Start break	Alert, always shown
Job fails	Couldn't convert: file is open in Word	Retry	Alert
Job finishes	Saved Essay.pdf	Open, Show in folder	Alert
Low battery	Controller battery 10%	None	Alert
Device connects	Headphones connected, 80%	None	Alert
Download finishes	report.docx downloaded	Convert, Upload, Shelf	Alert
Screenshot saved	Screenshot saved	Copy text, Upload, Shelf	Alert
Auto-Sort moves a file	Moved syllabus.pdf to School	Undo	Alert
Song changes	Title, Artist	Pause, Next	Alert, 3 s
Mic muted or unmuted	Muted	None	Alert, 1.5 s
Files copied in Explorer	2 files copied	Convert, Compress, Upload	Off

Rules that keep it calm:

- **One at a time.** Alerts queue in the order of the table above: mic first, song changes last.
- **Merging.** Alerts of the same type within 5 s merge, such as "3 files sorted".
- **Timing.** Alerts stay 4 s; mic, timer and failures stay 8 s. Hovering an Alert pauses its countdown.
- **Quiet times.** During fullscreen apps, Focus mode, presentations and Windows Do Not Disturb, only mic, timer and failure alerts show. Song changes are dropped; everything else waits.
- **Catch-up summary.** When quiet time ends, one Alert sums up what waited: "While you were busy: 2 downloads, 1 job done".

- **Pause.** Tray menu > Pause alerts for 1 hour or until tomorrow.

Media and Mixer

Ctrl+Alt+M opens one panel with two tabs, Now Playing and Mixer. Media works with any app that uses Windows' media controls: Spotify, YouTube in a browser, Apple Music, VLC and others.

Now Playing

- Album art, title, artist and the icon of the app playing it.
- Play or pause, previous, next, and a seek bar with elapsed and total time.
- Shuffle and repeat when the app supports them.
- When several apps are playing, a switcher row picks which one the controls drive.
- **In Peek:** the song title scrolls, and scrolling the mouse wheel over the pill changes the volume (system or the playing app, a setting).
- **Keys:** Space plays or pauses, Left and Right skip, Up and Down change that app's volume.

Mixer

- A row for every app making sound: icon, name, volume slider and mute button.
- A master volume slider, plus pickers for the default output and input devices.
- An output device per app, such as Spotify on speakers and Discord on the headset.
- **Remember volumes:** apps go back to their last level when they reopen (setting).
- **Keys:** Tab moves between apps, Left and Right adjust, M mutes.

Per-app output devices rely on a Windows interface that Microsoft does not document, so treat that feature as a stretch goal that may break with Windows updates.

Mic and Camera

Cove shows when any app is listening and gives you one mute key for all of them.

- **Live indicator.** A red dot appears in Peek whenever an app uses the mic, with an Alert when it starts: "Mic on: Discord". Hovering the dot lists every app using it. Cove reads this from the same usage records Windows' privacy settings use.

- **Global mute (Ctrl+Alt+A).** Mutes the default input device for every app at once. The Alert says "Muted" or "Live", and while muted the dot turns grey with a slash.
- **Talking while muted (stretch).** If your mic picks up speech while muted, an Alert says "You're muted".
- **Push-to-talk (stretch).** Hold a chosen key to unmute only while it is held.
- **Camera.** The same indicator for the webcam, off by default.

Devices and Battery

Cove announces devices as they connect and warns before they run out of battery.

- **Connect and disconnect alerts** for Bluetooth headphones, mice, keyboards and controllers: "Headphones connected, 80%".
- **Battery levels** come from what the device reports to Windows. Devices that report nothing appear without a percentage.
- **Low-battery warnings** at 20% and 10% by default, once per level per charge.
- **Devices panel (Cove key + B)** lists each connected device with its battery level, plus the laptop's battery, charging state and time left.
- **Settings:** which devices to watch and the warning levels.

Timer and Focus

Ctrl+Alt+T opens timers, Pomodoro and a stopwatch. The countdown stays visible in Peek the whole time.

- **Quick start:** buttons for 5, 10, 15, 25 and 45 minutes, plus a custom field. From the command bar: "timer 10" or "timer 1h30".
- **Pomodoro:** 25 minutes of work and 5 of break, four times, then a 15-minute break. All lengths can be changed.
- **Stopwatch:** start, lap, stop.
- **Several timers:** up to three at once, each with an optional label ("Laundry").
- **While running:** Peek shows "18:42". The panel has pause, resume, +5 min and stop.
- **When it ends:** an Alert that shows even in fullscreen, with an optional sound and +5 min or Start break.

Focus mode is a switch on any timer (or "focus 25"). Until the timer ends, it:

- Holds every non-urgent alert.

- Optionally turns on Windows Do Not Disturb.
- Optionally minimises apps you list (Discord, Steam) and shows a gentle reminder if you open one.
- Ends with a summary of what it held back.

Shelf

The Shelf holds files for a moment while you go somewhere else, then lets you drag them out where they belong. Ctrl+Alt+S opens it, and Peek shows how many files are on it.

- **Adding files:** the Files panel's Shelf action, dropping onto the Shelf panel, the Shelf button on download and screenshot alerts, right-click > Cove > Add to Shelf, or the "shelf" command.
- **References, not copies.** The Shelf remembers where each file is, so it uses no extra disk space. If a file moves or is deleted, its card says "Missing". An option copies files in instead, for downloads you plan to delete.
- **Dragging out:** drag one or several cards into any folder or app. This copies by default; hold Shift to move.
- **Acting on shelved files:** select cards to Convert, Compress or Upload them, with the same actions as the Files panel.
- **Clearing:** an X on each card and a Clear all button. Auto-clear after a restart or after 24 hours is a setting.

Download alerts and Auto-Sort

Cove watches your Downloads and Screenshots folders and suggests the next step for each new file. Auto-Sort can then file it away by your rules.

Knowing a download is finished. Cove ignores temporary files (.crdownload, .part, .tmp) and waits until the file's size has stopped changing for about 1 s and it can be opened. Only then does it show an Alert.

New file	Suggested actions
HEIC photo	Convert to JPG, Shelf
DOCX, PPTX or XLSX	Convert to PDF, Upload
Video over 25 MB	Compress, Upload

New file	Suggested actions
Screenshot	Copy text, Upload, Shelf
PDF	Upload, Shelf
ZIP	Extract here, Shelf
Anything else	Upload, Shelf, Show in folder

Auto-Sort is off until you create a rule. Each rule is a condition and a destination folder.

Example rule	Moves to
Type is PDF and name contains "syllabus" or "lab"	Documents\School
Extension is .exe or .msi	Downloads\Installers
Type is image	Pictures\Downloads
Came from a site containing "classroom"	Documents\School

- **Conditions:** file type, extension, name contains, size, and the site it came from when the browser records it.
- **Timing:** Auto-Sort waits until the download Alert is gone (up to 30 s), so you can still act on the file first.
- **Undo:** every move Alert has Undo. Settings keeps the last 100 moves, each with its own Undo.
- **Dry run:** shows what each rule would move, without moving anything.
- **Conflicts:** a file with the same name in the destination gets a number added, never a replacement.

Right-click menu

Selecting files in Explorer and right-clicking gives a Cove submenu. Its progress and results show in the pill like any other job.

- Cove
  - Convert to (formats for that file type)
  - Compress (Discord 10 MB, Email 25 MB, Custom)
  - Upload and copy link

- Copy text (images and PDFs)
- Add to Shelf
- Open in Cove (loads them into the Files panel)
- More (Rename, Strip metadata, Merge PDFs, Zip)

Several files become one job. The menu entry starts a small helper that passes the file paths to the running Cove, and starts Cove first if it isn't running.

Windows 11 has two right-click menus. Simple entries appear only under "Show more options". Appearing in the main, modern menu needs extra packaging work, so the plan is the classic menu first and the modern menu later.

Tray icon and Settings

The tray icon shows Cove's state at a glance: normal, a small dot while a job runs, or dimmed while alerts are paused. Left-click opens the Menu; right-click shows Files, Shelf, Timer, Pause alerts, Settings, Help and Quit.

Settings is a separate window with one page per area. Modules appear as PowerToys-style cards with an on/off switch and a one-line description.

Page	What you can set
General	Start with Windows, theme, language, updates, export and import settings
Pill	Monitor, distance from the top, size (S, M, L), motion (full or reduced), background, optional hover reveal with strip width and delay
Shortcuts	Record, change or turn off each shortcut; conflict warnings
Modules	On or off for each module
Alerts	Per event: Alert, Peek only or Off; duration; what may show during quiet times; catch-up summary
Files	Default formats and quality, output folder, naming, jobs at once, tool locations
Compress presets	Edit the built-in presets or add your own

Page	What you can set
Cloud	Sign in or out of OneDrive and Google Drive, default service, link access, upload folder
Watched folders and Auto-Sort	Folders to watch, rules, dry run, move history
Media and Mixer	Scroll-wheel behaviour, remembered volumes
Mic and Camera	Mute shortcut, indicator style, camera indicator
Devices	Devices to watch, warning levels
Timer and Focus	Presets, sounds, Pomodoro lengths, apps to minimise
Privacy	What Cove stores on your PC, clearing history

First run

The first launch teaches the Cove key, sets sensible defaults, and finishes in under a minute.

1. **Hello.** The pill drops down: "Hi, I'm Cove. Press Ctrl+Alt+Space anytime."
2. **Four-step tour inside the pill.** The Cove key; the Files panel ("select a file and press Ctrl+Alt+D"); automatic alerts; where Settings lives. Each step can be skipped.
3. **Choices.** Which modules to turn on (all by default except Auto-Sort, file-copied alerts and the camera indicator), and whether to start with Windows (on by default).
4. **Cloud sign-in.** Optional, and skippable until the first upload.
5. **Shortcut check.** Cove tries to register each shortcut and lists any conflicts with a suggested replacement.
6. **Tools check.** ffmpeg and ImageMagick ship inside Cove. If LibreOffice is missing, document-to-PDF shows "Install LibreOffice to enable" with a link, and everything else still works.

The tour can be replayed from Settings > General.

Everyday scenarios

Ten common moments, step by step, showing how the modules work together.

1. A phone photo that won't open (HEIC)

1. IMG_2041.heic lands in Downloads.
2. The pill drops down: "IMG_2041.heic downloaded" with Convert, Upload and Shelf.
3. Click Convert. It already suggests JPG, so press Enter.
4. "Saved IMG_2041.jpg" appears, and the new file sits next to the original.

2. A video too big for Discord

1. Select the 60 MB clip in Explorer and press Ctrl+Alt+D. The clip is already loaded.
2. Press 2 for Compress, choose Discord (10 MB), press Enter.
3. Close the panel if you like; Peek shows the spinner.
4. "Saved clip (compressed).mp4, copied" appears. Paste it into Discord with Ctrl+V.

3. Handing in an essay as a PDF link

1. Right-click Essay.docx > Cove > Convert to > PDF.
2. The pill shows progress, then "Saved Essay.pdf" with an Upload button.
3. Click Upload. "Link copied" appears; paste it into your submission.

4. Moving files between distant folders

1. Select three files and press Ctrl+Alt+S, or drop them on the Shelf.
2. Browse to the destination folder; the files wait on the Shelf.
3. Press Ctrl+Alt+S again and drag the files out into the folder.

5. Text from a lecture slide

1. Take a screenshot with Win+Shift+S and save it, or let Windows save it to Screenshots.
2. The pill shows "Screenshot saved" with Copy text.
3. Click Copy text, then paste into your notes.

6. A call on Discord

1. Join the call. A red dot appears in Peek and the Alert says "Mic on: Discord".
2. Press Ctrl+Alt+A to mute; the pill flashes "Muted" and the dot turns grey.
3. Press it again to speak. When the call ends, the dot disappears.

7. A study session

1. Ctrl+Alt+Space, then / and type "focus 25", then Enter.
2. Peek counts down. Alerts are held and Discord is minimised.
3. At 0:00, "Timer done" shows with Start break, followed by a summary of held alerts.

8. Gaming

1. Launch a fullscreen game. Cove stays completely hidden.
2. Downloads and song changes wait or are dropped.
3. If the controller drops to 10%, that warning still shows, if you allowed it.

9. A messy Downloads folder

1. In Settings > Auto-Sort, add rules such as "PDFs with 'lab' in the name go to School".
2. Turn on Dry run for a day to check what it would move.
3. Turn it on fully. Each move shows an Alert with Undo.

10. Music while you work

1. A song changes; the pill shows the title for 3 s.
2. Press Ctrl+Alt+M, then Right to skip, or scroll on Peek to lower the volume.
3. Esc closes the panel, and typing continues where you were.

Edge cases, safety and accessibility

These are the situations that separate a demo from an app people keep installed.

Screens and windows

- **Several monitors:** the pill shows on the main monitor, or on the monitor with the cursor (setting).
- **Display scaling:** sizes are set in logical pixels, so the pill looks the same at 100% to 200% scaling.
- **Fullscreen detection:** Cove asks Windows whether a fullscreen app, presentation or quiet time is active, rather than guessing from window sizes.
- **Apps run as administrator:** Windows blocks dragging from them into normal apps, so Cove's panel says "Can't receive files from an admin app" instead of failing silently.

Files and jobs

- **Free space:** checked before each job, with an estimate of the space needed.
- **Locked files:** named, with the app that holds them when Windows reports it.
- **Sleep and restart:** an interrupted job restarts when the PC wakes, or is marked failed and its partial file cleaned up.
- **Crash recovery:** if Cove crashes, a helper restarts it, and the Jobs list shows what was interrupted.

Safety and privacy

- Originals are never overwritten. Deletes go to the Recycle Bin. Strip metadata makes copies by default.
- Cloud sign-in tokens are kept in Windows Credential Manager, not in plain files.
- No data leaves the PC except files you choose to upload. No usage tracking by default.

Accessibility

- Every feature works from the keyboard alone.
- Screen readers get a name and role for every control.
- High-contrast themes, reduced motion and Windows text scaling are respected.

Performance targets (goals to measure, not promises): under 50 MB of memory and near 0% CPU when idle, and the pill opening within 100 ms of a shortcut.

Keyboard reference

All defaults in one place; every global shortcut can be changed in Settings > Shortcuts.

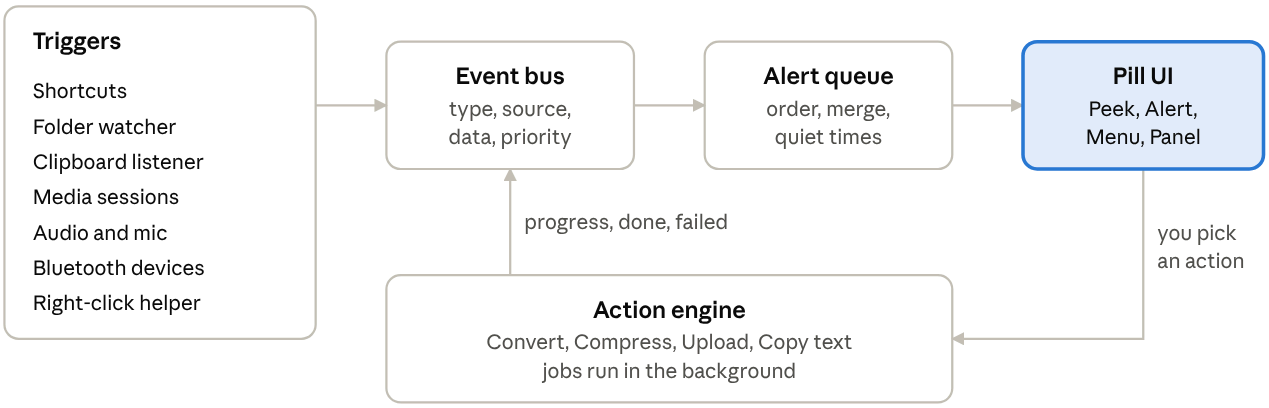
Where	Key	Does
Anywhere	Ctrl+Alt+Space	Cove key Menu
Anywhere	Ctrl+Alt+D	Files panel
Anywhere	Ctrl+Alt+M	Media and Mixer
Anywhere	Ctrl+Alt+T	Timer and Focus
Anywhere	Ctrl+Alt+S	Shelf
Anywhere	Ctrl+Alt+A	Mic mute or unmute

Where	Key	Does
Menu	D, M, T, S, B, J, A, comma	Files, Media, Timer, Shelf, Devices, Jobs, Mic, Settings
Menu	/	Command mode
Command mode	Tab, Enter, Up	Complete, run, last command
Any panel	Esc	Close (jobs keep running)
Any panel	Ctrl+P	Pin or unpin
Files panel	1 to 6	Convert, Compress, Upload, Copy text, Shelf, More
Files panel	Arrows, Enter	Choose an option, run
Files panel	Delete	Remove the selected file
Now Playing	Space, Left, Right, Up, Down	Play or pause, previous, next, volume
Mixer	Tab, Left, Right, M	Next app, adjust, mute
Timer	Space, +, Backspace	Pause or resume, add 5 min, stop
Shelf	Ctrl+A, Delete	Select all, remove
Alert	Ctrl+Alt+Space, then Enter	Runs the visible Alert's first button (Alerts never take focus)

How it works under the hood

Modules never talk to the pill directly: they publish events to one bus, so a new module shows up in the pill without changing the pill.

Every trigger passes through one event bus before reaching the pill



Cove architecture · triggers, event bus, alert queue, pill, action engine

Jobs report progress back through the same bus, which is why a running conversion appears in Peek even after its panel is closed.

Part	Windows feature or library
Global shortcuts	RegisterHotKey
Explorer selection	Shell windows interface (IShellWindows)
Copied files	Clipboard listener (AddClipboardFormatListener)
Download and screenshot watching	FileSystemWatcher
Now playing	Windows media session API (GlobalSystemMediaTransportControlsSessionManager)
Per-app volume	Core Audio sessions, through NAudio
Mic in use	Windows privacy usage records (CapabilityAccessManager)
Device battery	Windows Bluetooth and device APIs
Fullscreen and quiet times	SHQueryUserNotificationState
Window moves (hover option)	SetWinEventHook move and size events
Right-click helper to app	Named pipe

Part	Windows feature or library
Conversion	ffmpeg, ImageMagick, LibreOffice
OCR	Windows.Media.Ocr
Cloud	Microsoft Graph (OneDrive), Google Drive API
Sign-in tokens	Windows Credential Manager

Build order

Build in seven phases; after phase 4 you have a complete, demoable app, and everything later adds modules to a structure that already works.

1. **Core and Settings.** Tray icon, settings window with module cards, saving settings, start with Windows. Done when switches persist after a restart.
2. **The pill.** Hidden, Alert, Menu and Panel states with smooth animation, the Cove key, Esc, pinning, and focus handed back on close. Done when it opens in under 100 ms and never steals focus from typing.
3. **Files panel with Convert (images only).** Explorer selection, copied files, drag and drop, the job system with progress, done and failed states. Done when select, Ctrl+Alt+D, Enter produces a JPG.
4. **Media and download alerts.** Now Playing, song-change alerts, the Downloads watcher, the alert queue and quiet-time rules. This is the demo milestone.
5. **More file actions.** Video and audio conversion, Compress, Copy text, Shelf, the command bar.
6. **Cloud and Explorer.** Upload with OneDrive first, then Google Drive; the classic right-click menu.
7. **Remaining modules and polish.** Mixer, Mic and Camera, Devices and Battery, Timer and Focus, Auto-Sort, Bulk tools, first-run tour, accessibility pass.

Suggested stack: C# on .NET with WPF (or WinUI 3), the same family PowerToys uses. It reaches every Windows feature in this spec most directly.